Bangladesh University of Business and Technology
Department of Computer Science and Engineering

CSE 498A: Capstone Project/Thesis

PROPOSAL

Semester: Spring 2025

Title: Online Marks Entry System

Submission Date: 6th March, 2025

TEAM MEMBERS

Full ID	Full Name	Intake/Section

SUPERVISOR & COURSE TEACHER APPROVAL

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1. Problem Statement

Write your problem scenario in single paragraph format. For example:

Traditional methods of marks entry and management in educational institutions rely on manual processes, which are time-consuming, error-prone, and inefficient. Teachers often face difficulties in maintaining accurate student records, and students may experience delays in accessing their grades. The lack of an automated system can lead to discrepancies in grading, increased administrative workload, and challenges in data retrieval. An Online Marks Entry System [1] can address these issues by providing a digital platform for seamless marks entry, storage, and retrieval, ensuring accuracy, efficiency, and security in academic record-keeping.

2. Objectives

Write your objectives in list format. For Example:

a. *To develop a web-based system that allows teachers to input and manage student marks efficiently.*
b. *To implement role-based access control for teachers, students, and administrators.*
c. *To ensure data security, integrity, and accessibility for academic records.*
d. *To automate result calculations and generate real-time reports.*
e. *To provide students with instant access to their grades through a secure portal.*

3. Methodology

Describe your tentative proposed methodology with a workflow diagram. For Example:

*The Online Marks Entry System will be developed using a structured software development life cycle (SDLC) [2] approach which is shown in **Figure 1**, including:*

- *Requirement Analysis: Identifying key user requirements through stakeholder discussions.*
- *System Design: Creating database schemas, system architecture, and UI/UX wireframes.*
- *Development: Implementing the system using technologies like PHP, MySQL, HTML, CSS, and JavaScript.*
- *Testing & Debugging: Ensuring the system is free of errors through unit, integration, and user testing.*
- *Deployment & Maintenance: Hosting the system on a web server and ensuring ongoing updates and support.*

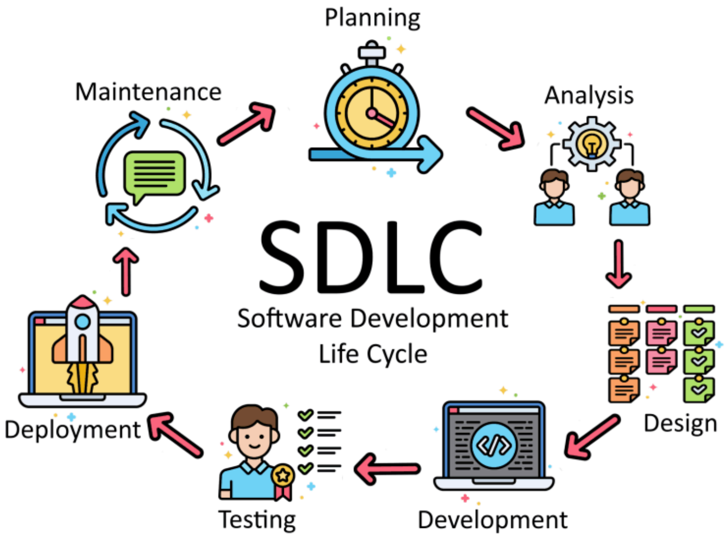

Figure 1. *SDLC process diagram*

4. Expected Outcomes

State your expected result or outcome from your project in list or paragraph format. For example:

i. *A fully functional Online Marks Entry System with role-based access.*
ii. *Reduced manual workload for teachers and improved accuracy in grade entry.*
iii. *Secure and centralized storage of student marks with real-time updates.*
iv. *Quick and easy retrieval of academic records for students and faculty.*
v. *Automated result generation, reducing administrative overhead.*

5. Complex Engineering Problem Mapping

Briefly list out the "K" (Knowledge Profile Attributes), "P"(Engineering Problems), and "A" (Engineering Activities) of complex engineering problem regarding your project. For Example:

Mapping with Knowledge Profile (WK)

- ***WK1 (Natural Sciences)****: Not directly applicable, as the project is more focused on software engineering and system design.*
- ***WK2 (Mathematics)****: Applicable in the context of automated result calculations and statistical analysis of student performance.*
- ***WK3 (Engineering Fundamentals)****: Essential for understanding the foundational principles of software engineering and system design.*
- ***WK4 (Specialist Knowledge)****: Required for implementing secure, role-based access control and database management.*
- ***WK5 (Engineering Design)****: Critical for designing the system architecture, database schemas, and user interfaces.*
- ***WK6 (Engineering Practice)****: Necessary for applying best practices in software development, testing, and deployment.*

- ***WK7 (Comprehension)**: Important for understanding the ethical implications of data security and the impact of the system on educational stakeholders.*
- ***WK8 (Research Literature)**: Useful for researching existing solutions, technologies, and methodologies to inform the system design.*

Mapping with Complex Engineering Problems (WP) [3]

- ***WP1 (Depth of knowledge required)**: The project requires in-depth knowledge of software engineering, database management, and web development.*
- ***WP2 (Range of conflicting requirements)**: Balancing the needs of teachers, students, and administrators while ensuring data security and system efficiency.*
- ***WP3 (Depth of analysis required)**: Requires abstract thinking to design a system that automates manual processes and ensures data integrity.*
- ***WP4 (Familiarity of issues)**: The issues of manual marks entry and data retrieval are common, but the solution requires a novel approach.*
- ***WP5 (Extent of applicable codes)**: The system must comply with data protection regulations and educational standards.*
- ***WP6 (Extent of stakeholder involvement & conflicting requirements)**: Involves multiple stakeholders with varying needs, such as teachers, students, and administrators.*
- ***WP7 (Interdependence)**: The system's components (database, user interface, security) are highly interdependent.*

Mapping with Complex Engineering Activities (EA)

- ***EA1 (Range of resources)**: The project involves the use of diverse resources, including software tools, databases, and human resources (developers, testers, and stakeholders).*
- ***EA2 (Level of interactions)**: Requires resolving significant problems arising from interactions between technical, security, and user experience issues.*
- ***EA3 (Innovation)**: Involves the creative use of engineering principles to design a system that automates and secures the marks entry process.*
- ***EA4 (Consequences to society and the environment)**: The system has significant consequences for educational institutions, improving efficiency and accuracy in academic record-keeping.*
- ***EA5 (Familiarity of issues)**: The project extends beyond previous experiences by applying principles-based approaches to solve common issues in educational administration.*

References

1. https://www.academia.edu/39042307/Online_marks_entry_system
2. https://aws.amazon.com/what-is/sdlc/
3. https://www.researchgate.net/publication/380839361_Implementation_of_complex_engineering_problem_solving_projects_in_a_Malaysian_engineering_programme